

AGASTYA REDDY KUKUDALA

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EDUCATION

University of Illinois Urbana-Champaign
B.S. Computer Science + Advertising

Urbana, IL
Expected December 2028

EXPERIENCE

Software Development Intern, Sketch Development

Feb. 2026 – May 2026

- Rebuilt 50 reusable HubSpot CMS modules across six pages using HubL, HTML/CSS, and JavaScript, resolving more than 70 rendering defects and enabling consistent service-page updates.
- Restructured service pages, added schema-rich FAQs, and created machine-readable service mappings, contributing to search impressions more than doubling from 75,000 to 200,000.
- Mapped application, database, network, compliance, and recovery dependencies for AWS migration planning so the team could identify prerequisites before implementation and account for business continuity requirements.

Software Development Mentorship, World Wide Technology (WWT)

Feb. 2025 – May 2025

- Led a five-person team with WWT engineer Michael Wilson to build a centralized scheduling platform used by three administrators and more than 10 Parkway clubs, replacing separate manual reminders; coordinated frontend, authentication, and backend handoffs.
- Built scheduling and event-management interfaces with JavaScript and HTML/CSS, integrating Python backend logic for user access and administrator approvals to connect student-facing workflows with administrative oversight.
- Connected the JavaScript frontend to Python backend logic for scheduling, user access, and approval workflows, bringing student-facing interactions and administrator oversight into one end-to-end release; coordinated UI, authentication, and backend handoffs.

Director of Advertising, Connect Me Tutoring

Jan. 2023 – Apr. 2026

- Built a continuously running, self-hosted Python/PostgreSQL system for a 13-person team to monitor media-outlet information, match stories to relevant outlets, and draft personalized LLM-assisted pitches, automating a manual workflow that previously required approximately 15 minutes per email.
- Implemented recipient exclusion lists, deduplication, human approval gates, and retry controls to prevent duplicate or unreviewed outreach while preserving editorial oversight.
- Helped secure more than 10 television placements through media outreach.

PROJECTS

ContextTree - Branching Conversation Engine for LLMs | Python, FastAPI, SQLite, React, TypeScript

- Built a branching chat application with React/TypeScript and FastAPI/SQLAlchemy, persisting conversation nodes in SQLite and reconstructing only the active root-to-node path for model prompts.
- Reduced estimated prompt tokens by 39.4% versus flat history, from 2,443 to 1,480 across nine requests in a synthetic branching benchmark.
- Added OpenAI and Z.ai model adapters and validated application behavior with 205 passing automated tests across the Python backend and React frontend.

AsterKV - Fault-Tolerant Distributed Key-Value Store | Go, gRPC, Protocol Buffers, Raft

- Implemented Raft leader election and write replication for a three-node Go key-value store, using gRPC/Protocol Buffers for communication and durable snapshots for recovery.
- Added checksummed persistent state, client-request deduplication, and compare-and-swap; observed no acknowledged-write loss or duplicate application of retried writes in the corresponding fault tests.
- Passed 70 repeated fault/safety test runs with Go's race detector enabled; restored writes within 0.40–1.70 seconds across five tests that force-killed the leader process.

InsiderPulse - SEC Insider-Purchase Analysis & Backtesting | Python, pandas, SQLite, requests, yfinance

- Built an EDGAR ingestion pipeline that parsed 1,892 SEC Form 4 filings without parse failures, extracting 66 open-market purchase records and grouping them into 47 signals.
- Implemented rule-based scoring by insider role, transaction size, and clustered buying; backtested available 7/30/90-day returns against SPY with look-ahead-bias checks.
- Stored transactions and evaluation results in SQLite, generated reports by signal characteristics and holding period, and added 92 automated tests with GitHub Actions CI.

TECHNICAL SKILLS

Programming Languages: C++, Python, Go, Java, SQL, JavaScript, TypeScript

Machine Learning & AI: PyTorch, Computer Vision, LLM APIs

Systems & Infrastructure: Distributed Systems, PostgreSQL, SQLite, gRPC, Protocol Buffers, REST APIs, Git, GitHub

Web Development: FastAPI, SQLAlchemy, React, Flask, HTML/CSS, HubL, HubSpot CMS

Testing & CI: pytest, GitHub Actions

AWARDS

FBLA: National Finalist (9th in U.S.) & 3x Missouri Financial Math Champion | USACO Gold | Mathleague National Finalist